# PAPER_1385 — UQFF Resolution of the Ehrenfest Rotating-Disk Paradox (CLOSED — Dispatch Whitepaper)

**Author:** Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Foundational / Special Relativity (CLOSED) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "ehrenfest_paradox"})` **Calculator surface:** `calculate_paradox({"paradox": "ehrenfest_paradox"})` **Closure helper:** `_l96_uqff_axiom_ehrenfest_paradox_closure()`

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## The Paradox

A rigid disk rotated to relativistic rim speeds has Lorentz-contracted circumference but uncontracted radius, implying $C/r < 2\pi$ — geometrically impossible for a rigid Euclidean disk.

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## UQFF Closed Identity

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Rim/axis chirality asymmetry = F_TRZ × β_i = 0.1 × 0.6029 = 0.0603
```

F_TRZ × β_i = 0.0603 supplies a rim-axis **chirality asymmetry** via the CW/CCW DPM grinding mechanism (PAPER_646). The disk is not rigid; rim and axis are different SCm/UA chirality channels.

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## Physical Interpretation

The rotating disk has two coupled chirality sectors — rim (CW-dominated) and axis (CCW-dominated). Their F_TRZ × β_i = 0.06 asymmetry absorbs the geometric mismatch without requiring rigid-body propagation.

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## Live Calculator Output

```python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "ehrenfest_paradox"})["value"]
```

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_ehrenfest_paradox_closure` in `uqff_pure_calculator.py` for the full field list.

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## NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

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## Reference

- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_ehrenfest_paradox_closure`
- Dispatch keys: `ehrenfest_paradox`, `ehrenfest`, `rotating_disk_paradox`
- Related foundational papers: PAPER_646 (F_U_Bi_i / F_U=1 ledger), PAPER_597 (negative-time dual existence), PAPER_1183 (paradox consolidation).

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